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Jacob W. Posey, Brayden Roque, Swagnik Guhathakurta, and Ryan W. Houim[a)]

AFFILIATIONS

Mechanical and Aerospace Engineering Department, University of Florida, Gainesville, Florida 32611, USA

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[a)]**Author to whom correspondence should be addressed:** rhouim@ufl.edu

ABSTRACT

A multidimensional numerical study was performed to explore the ignition and combustion mechanisms of an annular aluminum particle shell surrounding a trinitrotoluene (TNT) charge. The model equations consist of a fully compressible reacting gas coupled to a kinetic-theory-based Eulerian granular multiphase model. The parametric study explored the influence of thickness of the Al particle layer, initial Al packing, and diameter of the Al particles. The computed results show ignition and flame structures that are consistent with both delayed and prompt ignition of explosively dispersed reactive powder. In delayed ignition, the Al particles ignite several milliseconds after the dispersal when the particles interact with the surface of the TNT fireball during the negative phase of the blast. The resulting turbulent Al dust flame propagates from the inside toward the outside of the dispersed Al dust cloud. The results show that smaller-diameter Al particles ignite more quickly and propagate through the dispersed dust cloud more rapidly. Paradoxically, larger amounts of smaller Al particles are unburned after the turbulent flame propagates through the dust cloud in comparison to larger-diameter particles. Prompt ignition occurs when the Al particles ignite in the shock-heated air almost immediately as the particles start to disperse. The flame structure for prompt ignition scenarios is a non-premixed dust flame where the fuel and air are mixed by velocity slip between the gas and particles.

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I. INTRODUCTION

Metal particles have been used as additives to energetic materials and explosives due to their ability to increase performance over traditional formulations.[1,2] The burning timescales of these metallic particles are typically much longer than the detonation timescales of the energetic material. As a result, metal particles can be used to extend the duration of heat release of the explosive or to increase the strength of the primary shock if they react quickly. In cases where the particles burn too slowly to affect the primary shock, long duration afterburning of the metal particles can increase performance metrics, such as quasi-static overpressure.[3] However, these metal particles can only be effectively used if they ignite and burn during the post-detonation processes of the blast.

Figure 1 shows typical arrangements of reactive particles used in explosive charges. The mixed configuration places the metal particles in the explosive material, where they directly interact with the detonation wave. This mixed configuration has been studied extensively.[2,4,5] A different concept is to place the reactive particles in an annular shell surrounding a core of pure explosive material.[6,7] Expansion of the HE detonation products (DPs) disperses the particles radially outwards which may ignite to form a turbulent multiphase dust flame.[8]

A fundamental limitation of the loose powder configuration is related to the packing limit of granular material (a particle volume fraction ∼65% for random packing of spheres[9]). The packing limit restricts the reactive particle mass that can be placed in the annular gap around the HE core. Concepts that increase the mass of reactive particles involve replacing inert casing material with structural materials that are reactive.[10–14] These structural reactive materials are often manufactured by isostatic compression of reactive powders, such as Al, into solid slabs, cylinders, etc. Consolidation allows the maximum volume fraction of the reactive particles to be increased from ∼65% for loose fills to over 98% for consolidated reactive materials,[12] while, at the same time, reducing or even eliminating the inert mass that is typically used for the casing. These consolidated materials are typically brittle and often shatter into fine fragments after the HE charge is detonated.[10–12] The reactive powder produced from the shattered casing

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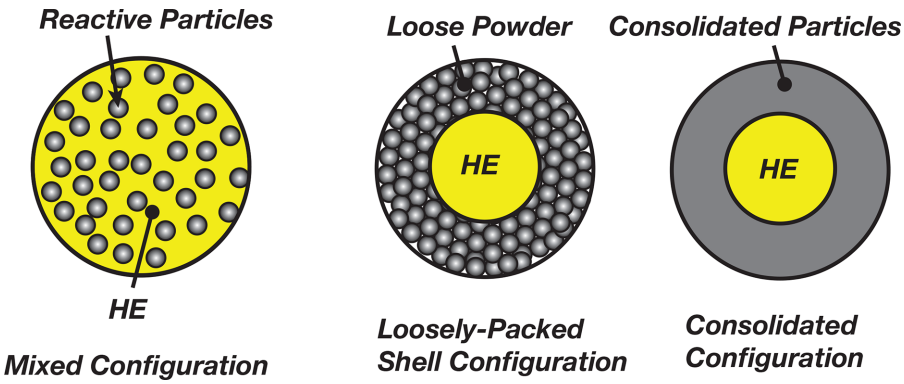


FIG. 1. Typical configurations of metal particle placement used in high-explosive (HE) charges.

material behaves similarly to loose powder, with the exception that the initial volume fraction is much higher.

Different modes of ignition and combustion for the loose-powder configuration have been observed depending on the diameter of the reactive particles and the size of the HE charge. The particle diameter controls the heating and burning timescales, while the HE charge diameter controls the gasdynamic timescales of the blast.[8] Frost *et al.*[8] found that charge configurations with small particles or large HE charges lead to prompt ignition of the reactive particles. In prompt ignition, the particles start burning almost immediately after the detonation of the HE charge and continue to burn as the particles are dispersed radially outward. The resulting flame structure resembles a growing white-hot fireball.[8] Delayed ignition occurs when the HE charge is smaller or the reactive particles are larger in diameter. In delayed ignition, the particles do not react immediately and, instead, they are dispersed by the HE charge to form a dust cloud. An ignition event produces a turbulent growing dust flame that propagates through the dispersed dust cloud. Figure 2 qualitatively shows the process of particle dispersal and combustion for delayed ignition. Scenarios where the particles do not ignite occur if the particles are too large or the HE charge too is small for the dispersed powder to absorb enough heat before gasdynamic expansion cools the detonation products and shock-heated air.

Fundamental understanding of the mechanisms that control prompt and delayed ignition scenarios is not well understood. The overall processes of the explosive dispersal and combustion of reactive powder are inherently multiphysical and involve coupling between multiphase flow, particle dynamics, gasdynamics, and combustion. For example, multiphase instabilities often produce large particle jets as they are dispersed by the expanding detonation products.[6,15] The overall role of these particle jets in the ignition and combustion dynamics of the explosively dispersed reactive powder has not been fully explored. Our earlier computational study suggests that the dispersal, ignition, and combustion of loose Al powder surrounding a trinitrotoluene (TNT) charge are tightly coupled.[16] These results suggest that multiphase flow effects, such as the Stokes number of the particles, may also have a large influence in addition to the heating and reaction timescales. Given the rich variety of mechanisms that multiphase flow, gasdynamic processes, and combustion may interact, it is likely that many other ignition and combustion modes of explosively dispersed Al powder exist.

Our earlier work explored the influence of particle diameter on the ignition and combustion of loosely packed Al dispersed by a TNT charge.[16] In this paper, we present results of two-dimensional axisymmetric numerical simulations that explore the influence of layer thickness, initial packing, and particle diameter on the ignition and

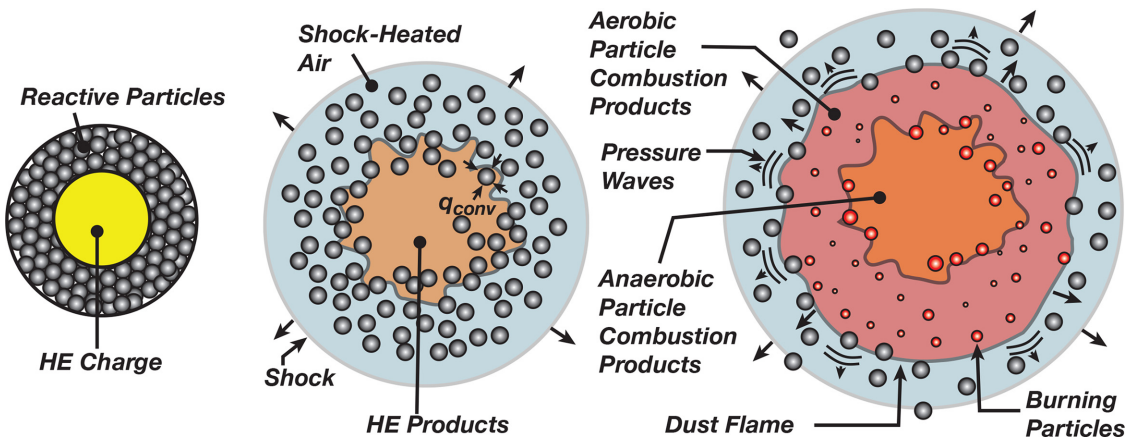


FIG. 2. Schematic diagram qualitatively showing the afterburning processes of an annular shell of loose reactive powder dispersed and ignited by a HE charge.

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combustion mechanisms of aluminum powder surrounding a TNT charge. The numerical code, HyBurn, is based on a recent Eulerian granular multiphase model that is fully coupled to a compressible reactive gas and includes the effects of particle collisions, compaction waves, and intergranular stress.[17] The geometric configuration of these simulations is shown in Fig. 3. Results that describe the ignition and combustion mechanisms of delayed and prompt ignition scenarios are discussed.

II. PHYSICAL MODEL AND NUMERICAL METHODS

The physical setup of the simulations consists of an annular shell of reactive particles surrounding a spherical high-explosive charge as shown in Fig. 3. The nominal input parameters for the simulations are listed in Table I. The explosive material considered is a 10 cm-diameter trinitrotoluene (TNT) charge. The geometrical configuration simulates the detonation of the TNT charge at the center of a 4 m-diameter by 4 m-high cylindrical chamber. The bottom side of the domain, located at the center of the charge, is assumed to be a symmetry plane. Rotational symmetry is assumed along the y-axis. The initial conditions inside the TNT are based on the constant-volume explosion combustion limit at 1.654 g/cc, 2870 K, and 9.089 GPa. The gas consists of four lumped species: TNT detonation products (DPs), air, TNT-air afterburning products (APs), and aluminum–air afterburning products (Al–AP) with specific heat polynomials from Kuhl and Khasainov.[18] The reactive particles are modeled as aluminum with a monodisperse size distribution, a density of 2700 kg/m^3, and a specific heat of 1176 J/kg K. Particle collisions were modeled as fully elastic with a coefficient of restitution (COR), e, of unity. Infinite-rate reactions are assumed to occur between the fuel-rich TNT detonation products and air.[7] The governing equations, aluminum particle combustion model, and numerical methods are discussed below.

The influence of Al particle diameter, layer thickness, and initial packing on the ignition and combustion is explored. The baseline case

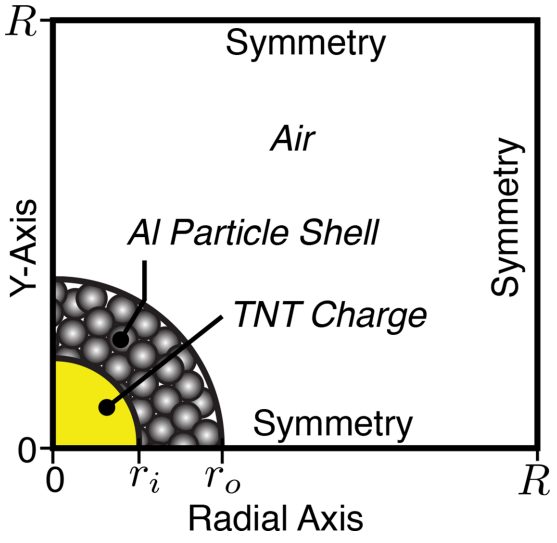


FIG. 3. Initial conditions and geometrical configuration for the two-dimensional simulations. Rotational symmetry about the y-axis is assumed. The domain is 2×2 m^2 in size, which models a 4 m radius and 4 m tall cylindrical blast chamber. Symmetry is assumed on the lower, upper, and right boundaries.

TABLE I. Initial and input parameters for the gas, particles, and explosive.

	Parameter	Value
Explosive	Material	TNT
	Diameter	10 cm
	Density	1654 kg/m^3
Gas	Initial pressure	9.089 GPa
	Initial temperature	2870 K
	Ambient temperature	298 K
	Ambient pressure	1 bar
Particles	Material	Al
	Density	2700 kg/m^3
	Specific heat	1176 J/kg K
	COR (e)	1
	Diameter	5 μm (Baseline)
	Layer thickness	5 cm (Baseline)
	Initial packing	50% (Baseline)
	Packing limit	65% (Baseline)

corresponds to a loose fill of Al particles with a monodisperse diameter of 5 μm, a layer thickness of 5 cm, and an initial packing (particle volume fraction) of 50%. Cases that correspond to shattered consolidated Al powder were simulated by increasing the initial particle volume fraction and packing limit beyond traditional values for packed rigid spheres. The simulations presented here neglect the shattering process of the consolidated material, which is a complex research topic by itself. Table II lists the parameters varied for all 11 cases considered as well as the resulting ignition mode.

A. Governing equations

The governing equations for this work model the gas and granular particles using Eulerian descriptions.[17] The model considers two sets of coupled governing equations, one for the reacting gas and another for the aluminum particles. The density of the aluminum particles is assumed to be constant. The governing equations for the gas phase are

$$\frac{\partial \alpha_g \rho_g Y_{g,i}}{\partial t} + \nabla \cdot (\alpha_g \rho_g Y_{g,i} \mathbf{v}_g) = \alpha_g \dot{\omega}_{g,i} + \dot{m}_{g,i}, \tag{1}$$

$$\frac{\partial \alpha_g \rho_g \mathbf{v}_g}{\partial t} + \nabla \cdot (\alpha_g \rho_g \mathbf{v}_g \mathbf{v}_g) = -\alpha_g \nabla p_g - \mathbf{f}_D + \mathbf{v}_s \dot{M}_{Al}, \tag{2}$$

$$\frac{\partial \alpha_g \rho_g E_g}{\partial t} + \nabla \cdot \left[\alpha_g \mathbf{v}_g (\rho_g E_g + p_g)\right] = -p_g \nabla \cdot (\alpha_s \mathbf{v}_s) - q_c + \phi_{visc} - \mathbf{f}_D \cdot \mathbf{v}_s + e_s \dot{M}_{Al}, \tag{3}$$

where α_g, $Y_{g,i}$, ρ_g, p_g, T_g, E_g, and $\mathbf{v}_g$ are the number of gas-phase species, volume fraction, mass fraction of gas-phase species i, density, pressure, temperature, total energy, and velocity vector for the gas phase, respectively. The homogeneous reaction rate of species i due to chemical reactions is denoted by $\dot{\omega}_i$. The net rate of phase change from aluminum particles to the gas-phase is represented by $\dot{M}_{Al}$ and the mass production rate of species i due to phase change is denoted by $\dot{m}_{g,i}$. The terms $\mathbf{f}_D$ and q_c represent the drag forces and convective heat transfer between the gas and granular particles. Conversion of

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TABLE II. List of parameters varied for the simulations.

Case	Layer thickness (cm)	Initial packing in layer	Particle diameter (μm)	Packing limit	Ignition mode
1	5.0	50%	5.0	65%	Delayed
2	1.0	50%	5.0	65%	Prompt
3	20.0	50%	5.0	65%	Delayed
4	5.0	25%	5.0	65%	Delayed
5	5.0	75%	5.0	99.5%	Delayed
6	5.0	95%	5.0	99.5%	Delayed
7	5.0	98%	5.0	99.5%	Delayed
8	5.0	50%	2.0	65%	Delayed
9	5.0	50%	3.0	65%	Delayed
10	5.0	50%	10.0	65%	Delayed
11	5.0	50%	20.0	65%	Delayed

random granular kinetic energy to gas-phase sensible energy due to viscous effects is denoted by ϕ_{visc}. The gas-phase total energy is given by

$$E_g = e_g + \frac{1}{2}\mathbf{v}_g \cdot \mathbf{v}_g, \tag{4}$$

where the gas-phase internal energy is

$$e_g(T_g, Y_{g,i}) = \sum Y_{g,i} e_i(T_g). \tag{5}$$

Here, e_i is the sensible energy for gas-phase species i where is computed using piecewise parabolas with coefficients available in Refs. 7 and 18.

A modified Jones–Wilkins–Lee (JWL) equation of state (EOS) is used for the gas,[7]

$$p(\rho, T, Y_{g,i}) = A\left(1 - \frac{\omega\rho}{R_1\rho_o}\right)\exp\left(\frac{-R_1\rho_o}{\rho}\right) + B\left(1 - \frac{\omega\rho}{R_2\rho_o}\right)\exp\left(\frac{-R_2\rho_o}{\rho}\right) + \rho R_m T, \tag{6}$$

with fitting parameters given in given in Table III. The gas constant of the mixture, R_{m}, is

$$R_{\text{m}} = R_u \sum \frac{Y_{g,i}}{M_{g,i}}, \tag{7}$$

where R_u is the universal gas constant and $M_{g,i}$ is the molecular weight of gas-phase species i. The molecular weights for each species are given in Table IV.

The gas-phase viscosity of the mixture, needed for particle drag and convective heat transfer, is[7]

$$\mu_g = \sqrt{\frac{T_g}{273\,\text{K}}} \sum X_{g,i}\mu_{0,i}, \tag{8}$$

TABLE III. JWL parameters for TNT.[7]

ρ_o (kg/m^3)	A (GPa)	B (GPa)	R_1	R_2	ω
1654	1576	43.24	6.662	2.158	0.4216

where $X_{g,i}$ and $\mu_{0,i}$ are the mole fraction and reference viscosity of gas-phase species i. Similarly, the thermal conductivity of the gas-phase mixture is[7]

$$\lambda_g = \sqrt{\frac{T_g}{273\,\text{K}}} \sum X_{g,i}\lambda_{0,i}, \tag{9}$$

where $\lambda_{0,i}$ is the reference thermal conductivity for species i. The reference viscosity and thermal conductivity for each species are given in Table IV.

A full description of the governing equations for the granular particles can be found in our earlier work.[17] The governing equations for the aluminum particles conserving mass, momentum, pseudo-thermal energy (PTE), which is the random translational kinetic energy of the particles, and sensible energy are

$$\frac{\partial \alpha_s\rho_s}{\partial t} + \nabla \cdot (\alpha_s\rho_s\mathbf{v}_s) = -\dot{M}_{Al}, \tag{10}$$

$$\frac{\partial \alpha_s\rho_s\mathbf{v}_s}{\partial t} + \nabla \cdot (\alpha_s\rho_s\mathbf{v}_s\mathbf{v}_s) + \nabla p_s + \nabla p_{\text{fric}} = -\alpha_s\nabla p_g + \mathbf{f}_{\text{D}} - \mathbf{v}_s\dot{M}_{Al}, \tag{11}$$

$$\frac{\partial \alpha_s\rho_s E_s}{\partial t} + \nabla \cdot (\alpha_s\rho_s E_s\mathbf{v}_s) = -p_s\nabla \cdot \mathbf{v}_s - \dot{\gamma} - \phi_{\text{visc}} - E_s\dot{M}_{Al}, \tag{12}$$

$$\frac{\partial \alpha_s\rho_s e_s}{\partial t} + \nabla \cdot (\alpha_s\rho_s e_s\mathbf{v}_s) = q_{\text{c}} - e_s\dot{M}_{Al} + \dot{\gamma}, \tag{13}$$

where α_s, ρ_s, T_s, and $\mathbf{v}_s$ are the volume fraction, density, temperature, and velocity vector for the aluminum, respectively. The parameters p_s

TABLE IV. Molecular weights, reference viscosity, and reference thermal conductivity for each gas-phase species: TNT detonation products (DP), TNT-air afterburning products (AP), air, and aluminum–air combustion products (Al–AP).[7,18]

Species	$M_{g,i}$ (kg/kmol)	μ_0 (Pa s)	λ_0 (W/m K)
DP	26.9	1.245×10^{-5}	0.020 96
AP	26.65	1.53×10^{-5}	0.0186
Air	28.85	1.716×10^{-5}	0.024
Al–AP	40.78	1.716×10^{-5}	0.024

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represent solids' pressure from particle–particle interactions, p_{fric} represents frictional pressure at high granular packing, E_s is pseudo-thermal energy (PTE) which represents random translational kinetic energy of the particles, and e_s is the internal energy of the particles. The term $\dot{\gamma}$ dissipates E_s from inelastic particle collisions. The term ϕ_{visc} is the dissipation of E_s due to viscous drag forces. Particle–particle interactions are modeled with constitutive relations that describe an intergranular stress with p_s and p_{fric} are described below.

The internal energy of the particles is computed assuming constant specific heat for the Al particles,

$$e_s = h_f^0 + C_v(T - T_{ref}), \tag{14}$$

where $C_v = 1177$ J/kg K, $T_{ref} = 298$ K, and $h_f^0 = 0$ J/kmol. The particle collisional pressure is given by[19]

$$p_s = \rho_s \Theta_s[\alpha_s(1 + 2(1+e)\alpha_s g_0)], \tag{15}$$

where g_0 is the radial distribution function, e is the coefficient of restitution (COR) (unity for this work), and Θ_s is the granular temperature defined by[19]

$$\Theta_s = \frac{2}{3}E_s. \tag{16}$$

The radial distribution function, g_0, is defined by[19]

$$\frac{1}{g_0} = 1 - \left(\frac{\alpha_s}{\alpha_{s,\max}}\right)^{1/3}, \tag{17}$$

where $\alpha_{s,\max}$ is the packing limit of the Al particles.

The frictional-collisional pressure is used in highly packed granular regions,[20]

$$p_{\text{fric}} = \begin{cases} 0 & \text{if } \alpha_s < \alpha_{s,\text{crit}}, \\ 0.1\alpha_s \dfrac{(\alpha_s - \alpha_{s,\text{crit}})^2}{(\alpha_{s,\max} - \alpha_s)^5} & \text{if } \alpha_s \geq \alpha_{s,\text{crit}}, \end{cases} \tag{18}$$

where $\alpha_{s,\text{crit}}$ is 0.5 and p_{fric} is in units of Pa. Further details can be found in Ref. 17.

The drag force in this work is given by the Gidaspow correlation,[19]

$$\mathbf{f}_{\text{D}} = K_{sg}(\mathbf{v}_g - \mathbf{v}_s), \tag{19}$$

where

$$K_{sg} = \begin{cases} 0.75 C_d \dfrac{\rho_g \alpha_g \alpha_s |\boldsymbol{v}_g - \boldsymbol{v}_s|}{d_s \alpha_g^{2.65}} & \text{if } \alpha_g \geq 0.8, \\ 150 \dfrac{\alpha_s^2 \mu_g}{\alpha_g d_s^2} + 1.75 \dfrac{\rho_g \alpha_s |\boldsymbol{v}_g - \boldsymbol{v}_s|}{d_s} & \text{if } \alpha_g < 0.8, \end{cases} \tag{20}$$

d_s is the diameter of the aluminum particles, and the drag coefficient, C_d, is

$$C_d = \begin{cases} 24 \dfrac{[1 + 0.15(\alpha_g Re)^{0.687}]}{\alpha_g Re} & \text{if } \alpha_g Re < 1000, \\ 0.44 & \text{if } \alpha_g Re \geq 1000, \end{cases} \tag{21}$$

and the Reynolds number is defined by

$$Re = \frac{\rho_g |\mathbf{v}_g - \mathbf{v}_s| d_s}{\mu_g}, \tag{22}$$

where μ_g is the gas-phase viscosity. The model for viscous damping of PTE, ϕ_{visc}, is adopted from Gidaspow,[19]

$$\phi_{\text{visc}} = 3K_{sg}\Theta. \tag{23}$$

The rate of convective heat transfer between the gas and particles is

$$q_{\text{conv}} = h_{sg}(T_g - T_s), \tag{24}$$

where the heat transfer coefficient, h_{sg}, is

$$h_{sg} = 6\frac{\alpha_s \lambda_g Nu}{d_s^2}, \tag{25}$$

where λ_g is the gas-phase thermal conductivity. The Nusselt number correlation of Gunn[21] is used

$$Nu = (7 - 10\alpha_g + 5\alpha_g^2)(1 + 0.7Re^{0.2}Pr_g^{1/3}) + (1.33 - 2.4\alpha_g + 1.2\alpha_g^2)Re^{0.7}Pr_g^{1/3},$$

and Pr_g is the gas-phase Prandtl number.

The granular dissipation term, $\dot{\gamma}$, converts E_s into e_s from inelastic collisions between particles,[22]

$$\dot{\gamma} = \frac{12(1-e^2)g_0\alpha_s^2}{\sqrt{\pi}} \frac{\rho_s \Theta_s^{3/2}}{d_s}. \tag{26}$$

B. Aluminum and TNT detonation product afterburning models

Afterburning of the fuel-rich TNT detonation products with air is modeled as an infinite rate reaction,[7,18]

$$1\,\text{kg DP} + \nu_{TNT}\,\text{kg Air} \rightarrow (1 + \nu_{TNT})\,\text{kg AP}, \tag{27}$$

where DP is fuel-rich TNT detonation products, AP is the DP-air afterburning products, and the stoichiometric mass air–fuel ratio is $\nu_{TNT} = 3.35$.

The aluminum afterburning reaction on a mass basis is[7]

$$1\,\text{kg Al} + \nu_{Al}\,\text{kg Air} \rightarrow (1 + \nu_{Al})\,\text{kg Al} - \text{AP}, \tag{28}$$

where Al–AP are the aluminum–air afterburning products and $\nu_{Al} = 4.03$.

The burning rate model for the aluminum particles is based on a classical resistor approach that combines diffusion-limited and chemical-kinetic limited processes.[23–25] The mass consumption rate of the Al particle mass needed in the governing equations [see Eq. (10)] is

$$\dot{M}_{Al} = N_d A_s \frac{k_d k_k}{k_d + k_k}\rho_{ox}, \tag{29}$$

where N_d, A_s, and ρ_{ox} are the number density, surface area, and oxidizer mass density at the particle surface, respectively. The diffusion-limited and kinetic-limited rate constants are denoted by k_d and k_k, respectively. Assuming monodisperse particles,

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$$N_d A_s = 3\frac{\alpha_s}{r_s}, \tag{30}$$

where r_s is the Al particle radius.

Basic theory is used to estimate k_d and k_k, as is often traditionally done for coal and carbon particle combustion.[26] However, combustion of aluminum particles is complicated by phenomena such as a growing oxide shell that may fracture, diffusion through a porous oxide shell, growing oxide lobes, or even micro-explosions.[27–30] These phenomena make it difficult to directly use theoretically based classical expressions for k_d and k_k. Through algebraic manipulation, it is possible to transform the rate-constants in Eq. (29) into an equivalent form based on burning times,

$$\dot{M}_{Al} = 3\frac{\alpha_s \rho_s}{t_t} = 3\frac{\alpha_s \rho_s}{t_d + t_k}, \tag{31}$$

where t_t, t_d, and t_k are total, diffusion-limited, and kinetic-limited burning times of aluminum particles, respectively. Equation (31) allows the direct use of burning time data or correlations. Data used to correlate t_d must be based on Al burning particles that are known to be in a purely diffusion-limited regime. Similarly, data that are used to correlate t_k must be based on burning times of Al particles that are known to be in a kinetic-limited regime. Fortunately, there is an abundance of burning time data in both regimes for Al particles.[29,31–33]

The correlation of Beckstead[29] with a Ranz-Marshall correction[7,34–36] for convective mass transfer enhancement is used to compute the burning time in the diffusion-limited regime,

$$t_d(s) = \frac{C_1 d_s^{1.8}}{p^{0.1} T^{0.2} X_{O2}} \frac{2}{Nu_p}, \tag{32}$$

where X_{O2} is the mole fraction of O_2, $C_1 = 7.35 \times 10^{-6}$ s, and

$$Nu_p = 2 + 0.6 Pr_g^{1/3} \sqrt{Re}. \tag{33}$$

A burning time correlation[37–39] based on experimental burning times of nano-sized aluminum particles[31] is used to compute the kinetic-limited burning time,

$$t_k(s) = C_2 \left(\frac{d_s}{0.08}\right)^{0.3} \left(\frac{8}{p}\right)^{0.5} \frac{e^{\frac{T_a}{T_f}}}{X_{O2}^{0.4}}, \tag{34}$$

where $C_2 = 250 \times 10^{-9}$ s, $T_a = 9621$ K, and $T_f = (T_s + T_g)/2$ is the film temperature. The d_p, p, and T are in units of μm, atm, and K, respectively, in Eqs. (32) and (34). Figure 4 shows the burning time of Al particles as a function of diameter at 1 atm.

An ignition temperature, T_{ign}, is typically applied to the combustion of aluminum particles[7,29,40] to account for the influence of the protective aluminum oxide coating surrounding the Al particles. If an ignition temperature criterion is applied, we have

$$\dot{M}_{Al} = \begin{cases} 0 & T < T_{ign}, \\ 3\dfrac{\alpha_s \rho_s}{t_d + t_k} & T \geq T_{ign}. \end{cases} \tag{35}$$

However, in this work, we make the ansatz that pressures the Al particles are subjected to during the dispersal process is high enough to fracture the aluminum oxide coating. This would expose neat Al to the oxidizing environment and allow the particles to react well before the ignition temperature criterion is reached. Thus, we use $T_{ign} = 0$ K for this work, similar to Soo *et al.*[25] and Briand *et al.*[41] for Al particles and coal combustion.[26]

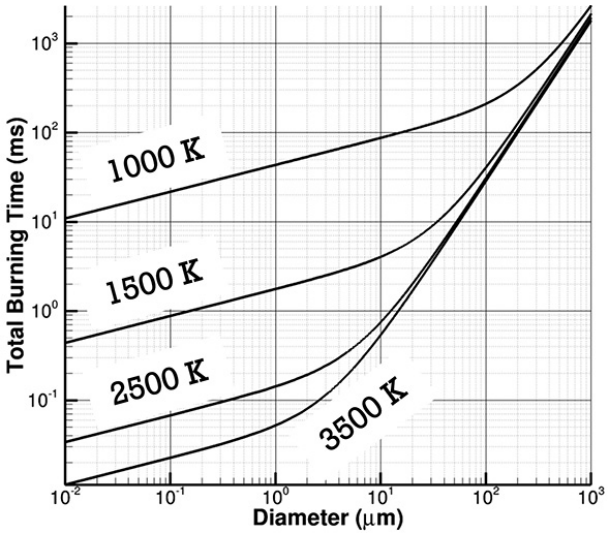


FIG. 4. Total burning time $t_t = t_k + t_d$ for aluminum particles at 1 atm as a function of d_s for several T.

The burning times of the Al particles are far longer than the duration of the simulations considered in this work when the temperature is ∼1000 K or less. The burning time of the Al particles becomes fast enough to strongly couple with the flow when the temperature is greater than 1500 K, which is close to the lower limit of some ignition temperature thresholds.[7,29,42]

C. Numerical methods

The governing equations are solved using an operator splitting method, separating the hydrodynamic terms and source terms. The hyperbolic terms are solved using a high-order Godunov-based method[17,43] with a third-order Runge–Kutta scheme[44] used for time integration. The fifth-order monotonic upstream-centered scheme for conservation laws (MUSCL) algorithm[45–47] and Harten, Lax, van Leer with contact (HLLC) flux[48] are used to compute the gas-phase fluxes. A modified advection upstream splitting method (AUSM) flux scheme is used for the Al particles.[17,49] Implicit large eddy simulation[50] (ILES) with a low-Mach correction[51] to the Riemann solver is used to model turbulence and turbulent mixing. This approach has been shown to be successful for a variety of turbulence and turbulent mixing applications including Richtmyer–Meshkov (RM) instabilities.[43,52,53]

The simulations were performed on an adaptive mesh using the AMReX library[54] to efficiently achieve adequate grid resolution. The simulations use four levels of refinement with a grid spacing of 0.98 mm at the finest resolution. The results of a grid independence study, shown in Fig. 5, indicate that the finest grid with four levels of refinement is sufficient to capture the bulk features of Al combustion including the timing and magnitude of the peak heat release rate as well as the duration and magnitude of sustained Al combustion. Note there are some differences in the heat release rate shown in Fig. 5 between 3 and 5 ms as the refinement level changes. However, the differences in the heat release rates are several orders of magnitude lower

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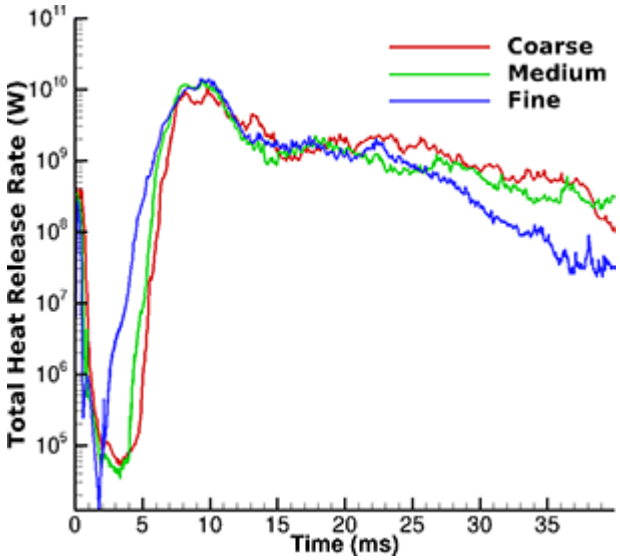


FIG. 5. Computed rate of heat release for case 1 with minimum grid sizes of 3.91 (coarse), 1.95 (medium), and 0.98 (fine) mm.

than the peak and periods of sustained energy release. Thus, these differences in heat release rate between 3 and 5 ms are small and do not affect the overall combustion dynamics. Examination of the temperature fields reveals that the overall timing and structure of the ignition and combustion processes of the dispersed Al are essentially the same at all resolutions.

D. Verification and validation

The HyBurn code used to perform the simulations in this paper has been extensively verified and validated against a variety of test problems. These verification problems are presented in our earlier work[17,43] and include a variety of Riemann problems, premixed laminar flames, detonations, dust-gas shock tube problems,[55] and granular shocks.[56] The granular multiphase flow model has been verified against experimental results for shock-particle curtain interactions,[17,57,58] transmitted oblique granular shocks in dust layers,[17,59] and dispersal of dust from a loose layer by a shock wave.[60,61]

The aluminum combustion model was verified by performing a numerical simulation of a cellular aluminum–air detonation.[24,37,38,41,62–64] Here, we consider a completely enclosed channel 30 m-long and 1.1 m-high channel filled with a mixture of air and Al particles at an equivalence ratio of 1.61. The diameter and mass loading of the Al particles are 13.5 μm and 0.5 kg/m^3, respectively. Symmetry boundary conditions are used for all surfaces in the channel. The governing equations were on an adaptive grid with six levels of refinement, giving an effective resolution of 2.1 mm.

The detonation is initiated by placing a series of high-pressure (200 atm) and high-temperature (2000 K) hot spots on the left end of the channel. The resulting detonation propagates to the right and becomes cellularly unstable. Measurements show that the detonation cell size for this Al–air mixture is ~40 cm.[62] Calculations using a similar hybrid kinetic-limited and diffusion-limited Al combustion model showed slightly irregular detonation cells of size 40 ± 10 cm.[41] The computed numerical smoke foil (trace of the maximum pressure) shown in Fig. 6(a) indicated that the present Al particle combustion model produces slightly irregular detonation cells of size 44 ± 15 cm, which is within the range of uncertainty of previous experimental and modeling results.[41,62] Figure 6(b) shows that the Chapman–Jouguet detonation velocity, D_{cj}, is within the instantaneous velocity of the leading shock.

III. RESULTS

A. Case 1: Baseline

A time series of temperature and Al particle volume fraction fields showing the dispersal, ignition, and combustion of the aluminum particles for case 1 is shown in Fig. 7. The dynamics of the dispersal, ignition, and combustion for case 1 are largely the same as in our previous work.[16] First, pressure and drag forces from the high-pressure gas push on the Al particle bed from the inside. These forces compact the particle bed while it accelerates outward. The outer edge of the densely compacted particle layer, in turn, acts like a porous piston that drives the air shock outside of the Al particle layer. The initial pressure forces are strong enough that the inner edge of the particle layer separates from the detonation products. Multiphase instabilities produce particle jets (or particle "fingers") that grow as the flow continues to expand.[6,15] The flow reverses during the negative phase of the blast, which is observed by the inner edge of the particles moving

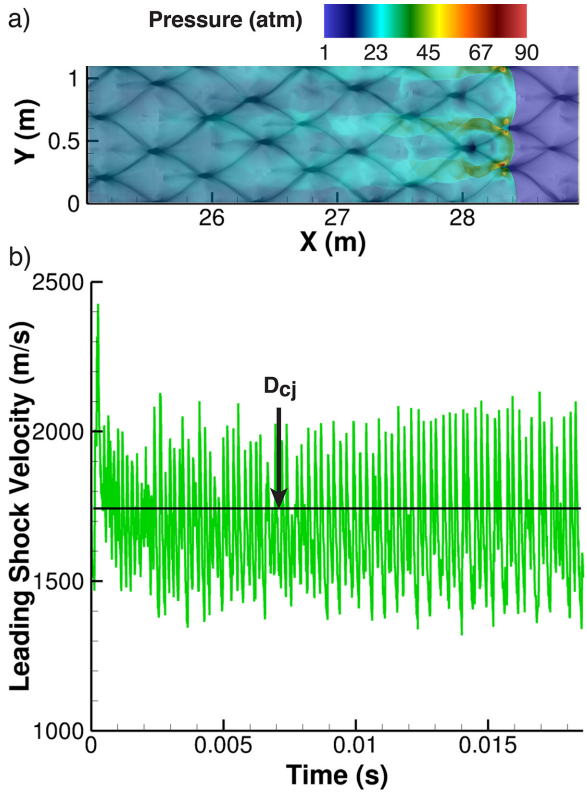


FIG. 6. Computed results for a cellular Al-air detonation with 13.5 μm-diameter particles and a mass loading of 0.5 kg/m^3 showing (a) a numerical smoke foil and pressure field and (b) the leading shock velocity as a function of time.

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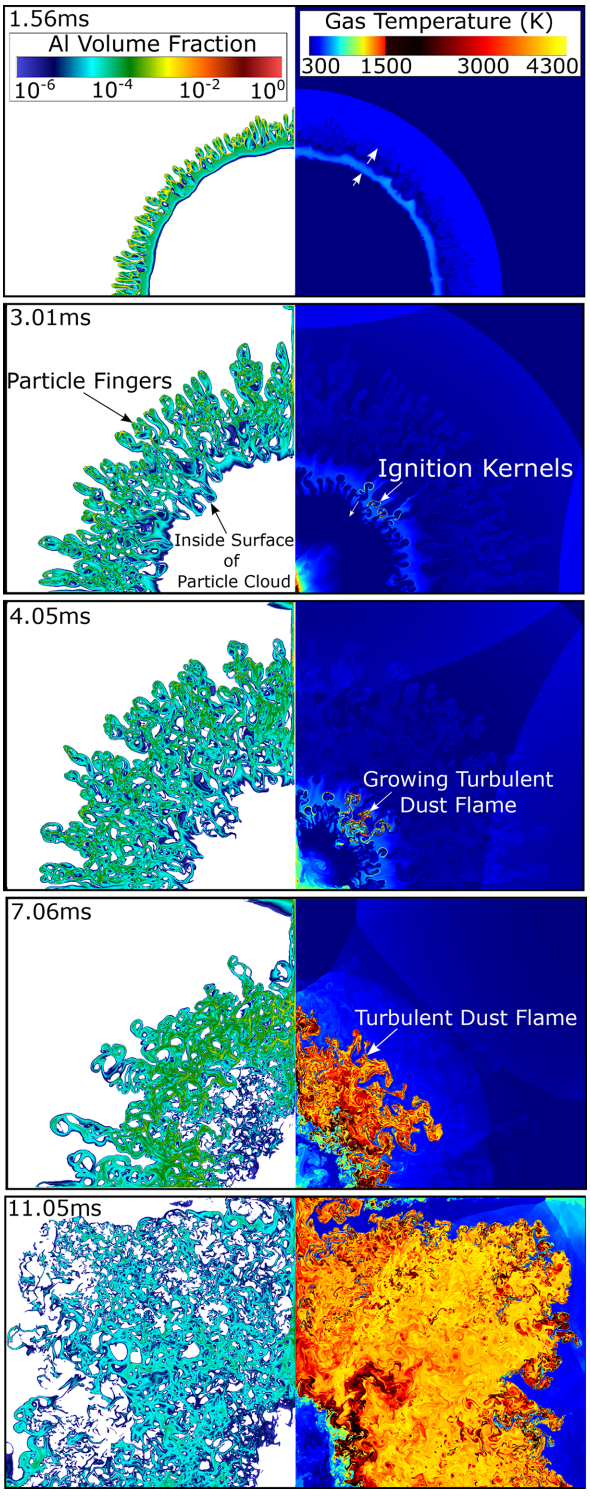


FIG. 7. Time sequence of (left) Al particle volume fraction and (right) gas-phase temperature for case 1. The time in ms is shown in the upper left-corner of each volume fraction map. The entire domain ($2 \times 2\,m^2$) is shown in each image. Multimedia view: https://doi.org/10.1063/5.0065312.1

toward the center of the charge. Inertia carries some of these inwardly moving particles into contact with the surface of the fireball. This ignites flame kernels at the inner edge of the Al dust cloud, which can be observed by localized regions where $T > 3500\,K$ at 3.01 and 4.05 ms. The resulting turbulent premixed Al dust flame rapidly propagates through and consumes the dispersed Al particles.

A position–time (X–t) diagram for case 1 is shown in Fig. 8. The X–t diagram is created by sampling temperature data along 17 rays and radially averaging the results. The X–t diagram shows the primary shock, secondary shock, negative phase of the blast, and ignition of the Al particle cloud. The rapid radial growth of the Al dust flame is clearly observed after 5 ms. The localized ignition kernels are more difficult to visualize on the X–t diagram due to the radial averaging of the individual rays. However, the ignition kernels appear before the secondary shock interacts with the Al dust cloud. Nevertheless, the interaction of the secondary shock has a substantial effect on promoting the growth of the Al flame by inducing Richtmyer–Meshkov (RM) instabilities.[52,65] The RM instabilities enhance turbulent mixing between the particles and air, which, in turn, enhances the flame propagation rate.

B. Effect of Al layer thickness: Cases 2 and 3

Position–time diagrams for cases 2 and 3, which examine the influence of Al particle layer thickness, are shown in Fig. 9. The X–t diagram for case 3, which has a 20 cm-thick Al particle layer, is qualitatively similar to the X–t diagram for case 1 (5 cm-thick layer). The dispersal, ignition, and combustion are similar to case 1, with the exception that the timescales are longer. The secondary shock interacts with the particles at about 4 ms for case 1, but the same interaction is delayed to about 9 ms for case 3. The thicker particle layer, which has higher Al mass, requires more time to accelerate, which, in turn, delays expansion of the high-pressure detonation products and formation of the secondary shock.

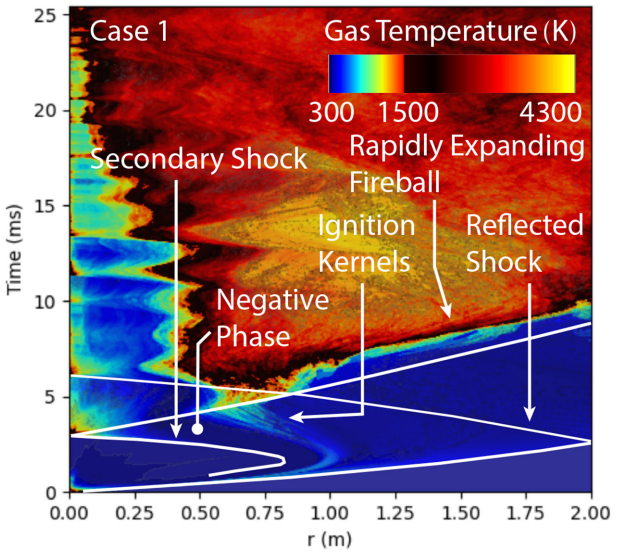


FIG. 8. Position–time diagram showing gas-phase temperature for case 1.

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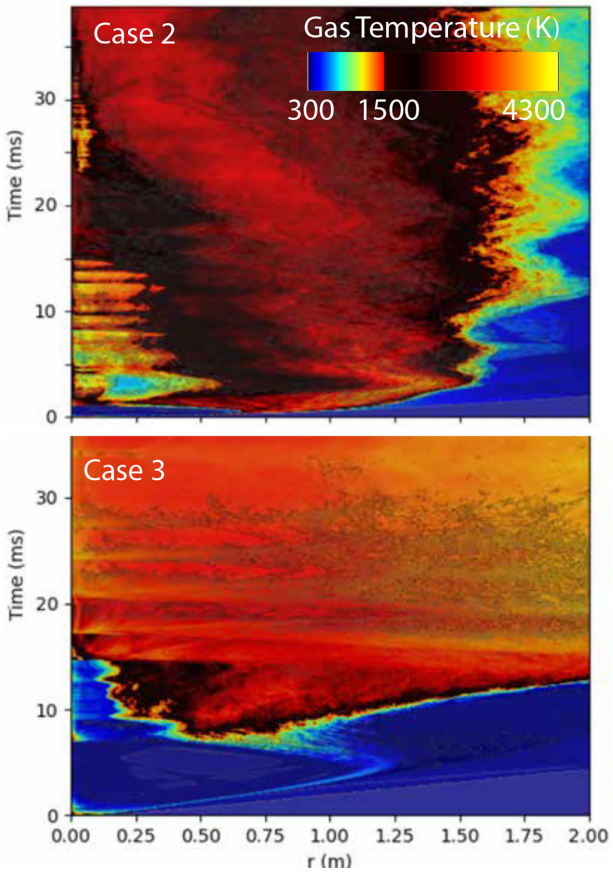


FIG. 9. Position–time diagram for cases 2 and 3 with aluminum particle layer thicknesses of 1 and 20 cm, respectively.

The X–t diagram for case 2 (1 cm-thick Al particle layer) shows dramatically different behavior than for cases 1 or 3. The time sequence of gas temperature and Al particle volume fraction shown in Fig. 10 indicates a completely different flame structure. Temperatures over 3500 K, which is indicative of Al burning, occur at very early times of 0.21 ms indicate rapid ignition of the particles during the initial expansion phase of the blast. Afterwards, the Al dust flame grows with the expansion of the flow behind the primary shock.

The heat release due to Al afterburing and normalized Al mass (M_{Al}/M^0_{Al}, where M^0_{Al} is the initial Al mass surrounding the HE core) is shown in Fig. 11. Cases 1 and 3 both show a delay in the ignition and the onset of rapid combustion (indicated by the sharp decrease in Al mass). The timings for ignition and the onset of rapid burning are delayed for cases 1 and 3. The peak heat release rate for case 3 occurs about 2 ms after the peak in case 1. Case 3 has much more Al mass; as a result, only about 30% of the Al is burned due to depletion of air.

The Al mass and heat release rates show different behavior for case 2. The peak of the heat release rate from Al combustion occurs very promptly, at ∼200 μs. In addition, the Al starts to deplete almost immediately. More than 60% of the Al has been burned for case 2 before the Al particles have even ignited in case 1.

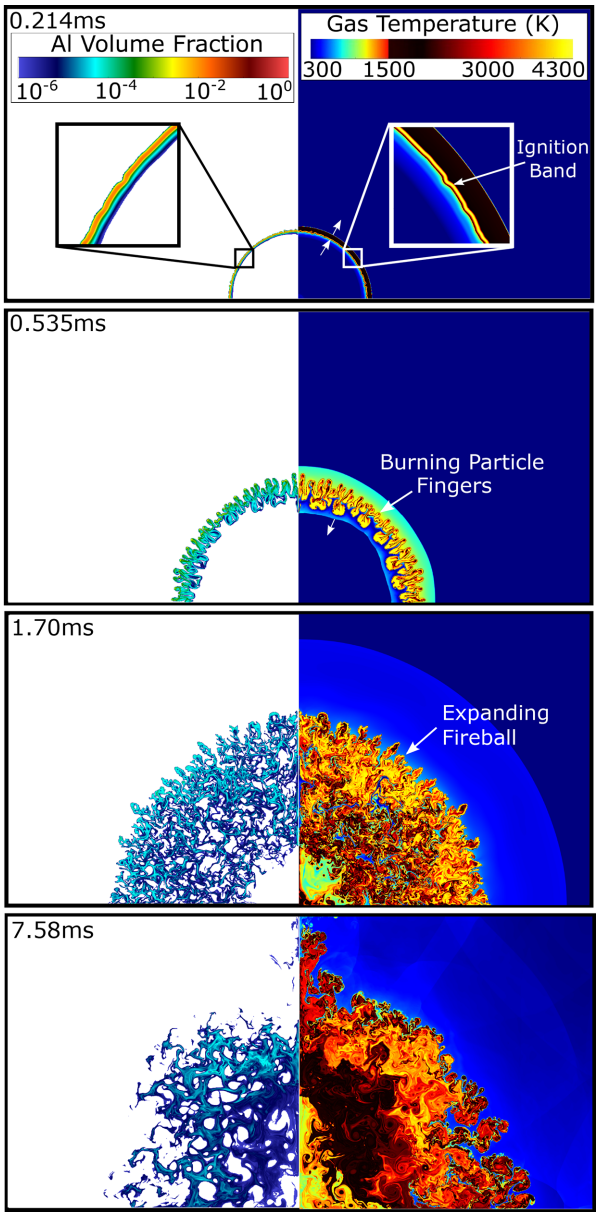


FIG. 10. Time sequence of (left) Al particle volume fraction and (right) gas-phase temperature for case 2. The time in ms is shown in the upper left-corner of each volume fraction map. The entire domain is shown in each image. Multimedia view: https://doi.org/10.1063/5.0065312.2

C. Effect of Al initial packing: Cases 4–7

Position–time diagrams for cases 4, 5 and 7, which examine the effect of initial Al volume fraction, are shown in Fig. 12. The X–t diagrams for all three of these cases are qualitatively similar to that of case 1. The Al particles ignite after a time delay after they disperse into a particle cloud. Ignition occurs on the inner edge of the dust cloud and the flame kernels grew into a turbulent dust flame for all of these cases. The timing of the secondary shock

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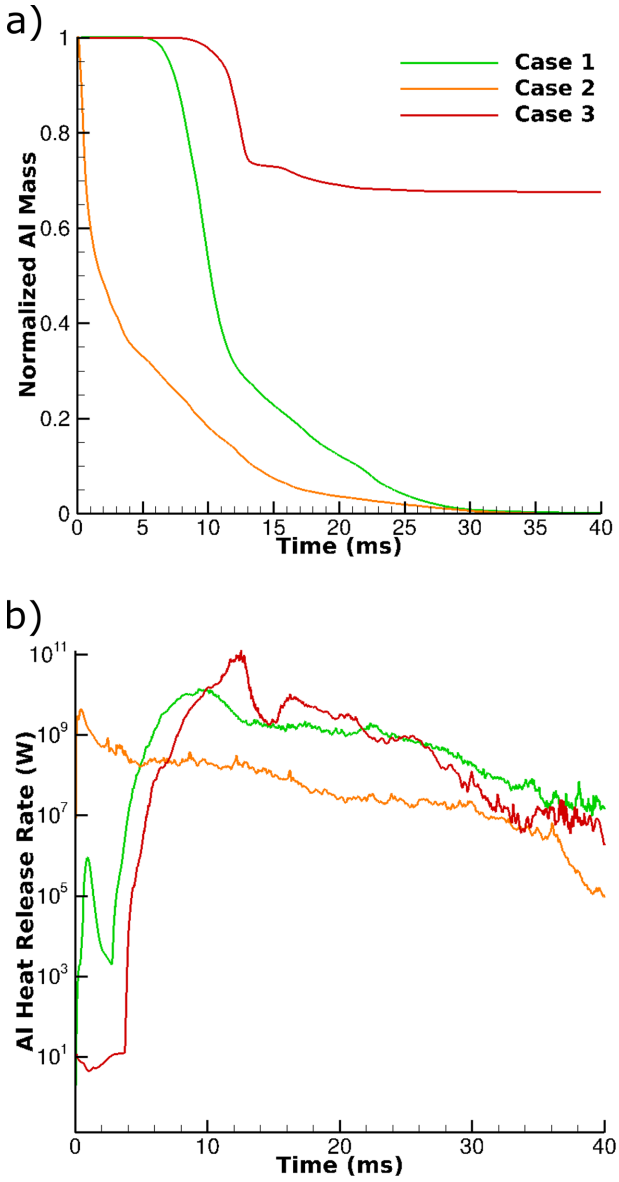


FIG. 11. Results for cases 1–3 showing (a) normalized Al mass (M_{Al}/M^0_{Al}) and (b) rate of heat release due to Al combustion.

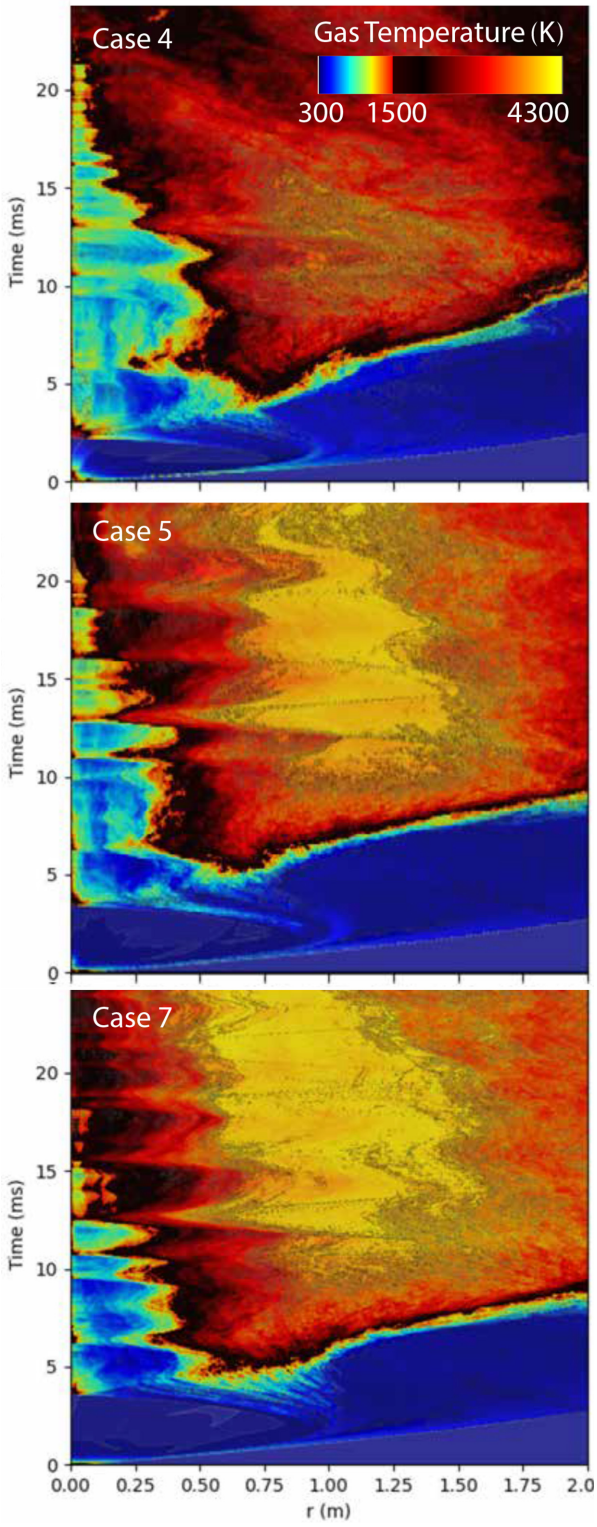


FIG. 12. Position–time diagram for cases 4, 5, and 7 with initial Al volume fractions of 25%, 75%, and 98%, respectively, in the Al layer.

interaction with the dispersed dust and onset of ignition occurs between 4 and 5 ms.

Figure 13 shows the normalized Al mass and heat release rate for cases 1 and 4–7. In all of these cases (same layer thickness and particle diameter), the rate of heat release peaks at almost ∼10 μs. The only exception is case 4, which considers a very dilute Al volume fraction of 25%, where the peak heat release rate occurs slightly before the other cases. This is more clearly shown by comparing the normalized Al mass for case 4 with respect to the other cases. The maximum rate of Al mass loss for case 4 occurs ∼1 ms before the other cases. After the

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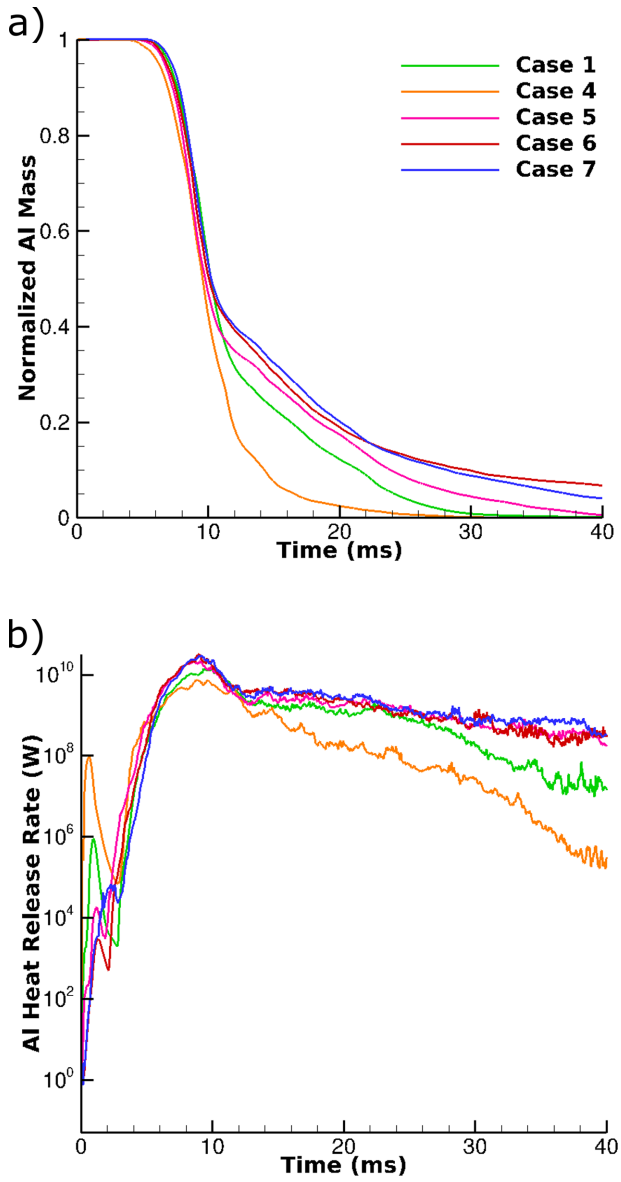


FIG. 13. Results for cases 4–7 showing (a) normalized Al mass (M_{Al}/M_{Al}^0) and (b) rate of heat release due to Al combustion.

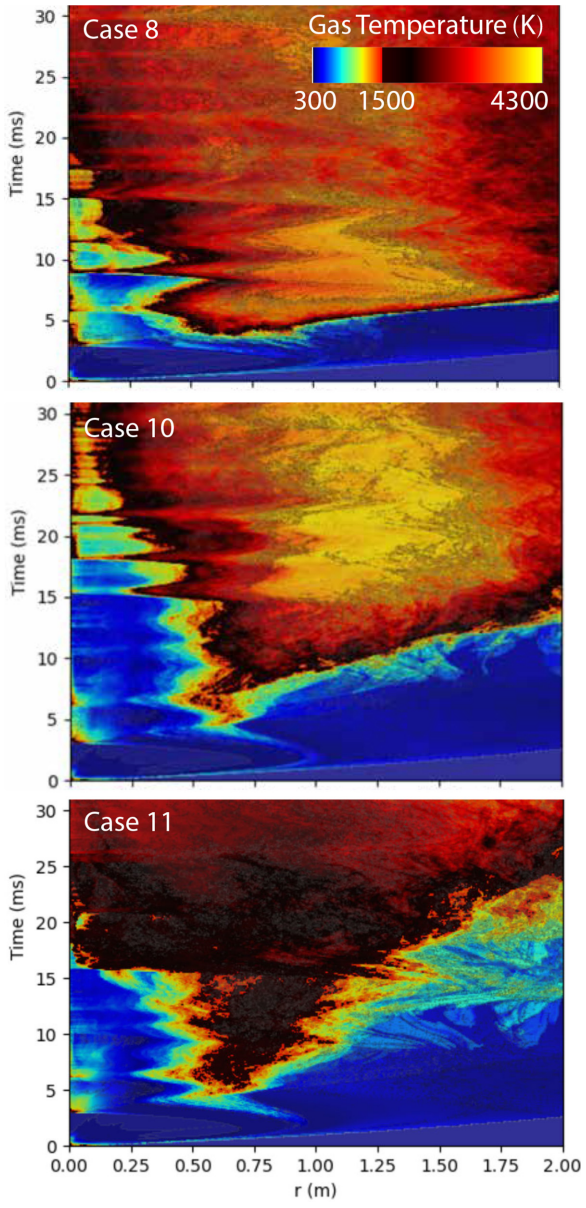


FIG. 14. Position–time diagram for cases 8, 10, and 11 with particle diameters of 2, 10, and 20 μm, respectively.

rate of heat release peaks, all of the cases follow essentially the same trend, with the exception of case 4. Case 4 has low Al mass, which completely depleted after 32 ms. These results indicate that the timing of the ignition and propagation of the Al dust flame are not strongly influenced by the initial packing of the Al particles, provided the initial volume fraction is in a physically realistic range (~50% or above).

D. Effect of initial particle diameter: Cases 8–11

Position–time diagrams for cases 8, 10, and 11 are shown in Fig. 14. The primary differences between these cases are related to the diameter effect on the expansion rate of the Al fireball after ignition. This can be observed by examining the slope of interface between the shock-heated air and high-temperature Al combustion products when the fireball rapidly grows. The Al dust flame for case 8 (2 μm-diameter particles) expands very quickly after ignition. The Al fireball grows much more slowly for the other cases. The flame for case 11 (20 μm-diameter particles) does not reach the outer radius of the combustion chamber until ~25 ms, which is 15 to 20 ms longer than case 8.

Figure 15 shows the normalized Al mass and heat release rate for cases 1 and 8–11. The Al heat release rate shows similar ignition and flame expansion trends as Fig. 14. The Al heat release rate shows that

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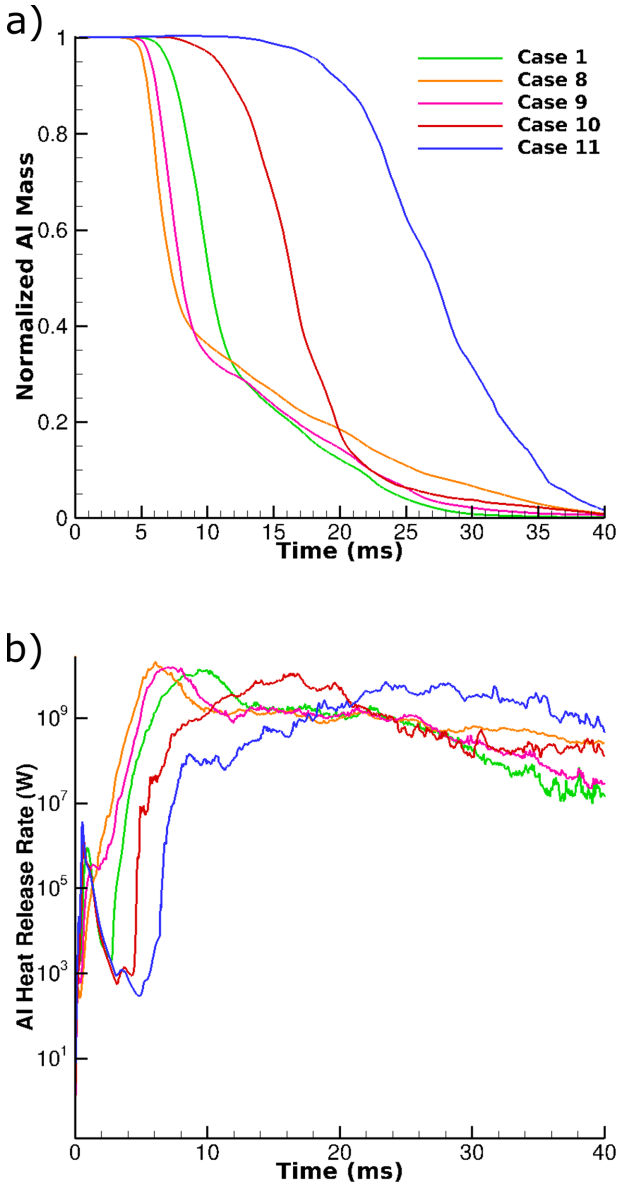


FIG. 15. Results for cases 8–11 showing (a) normalized Al mass (M_{Al}/M^0_{Al}) and (b) rate of heat release due to Al combustion.

ignition occurs in order of increasing particle diameter and that the duration of rapid Al depletion occurs over longer timescales with increasing particle diameter. The peak heat release rate occurs earlier and is larger in magnitude for the cases with smaller with Al particle diameters. These trends are largely due to the diameter effect on the Al particle burning time shown in Fig. 4. The ratio of burning times for the 20 and 2 μm-diameter particles is roughly 6. Thus, not only do the larger particles have much longer heating timescales, but they also take much longer to burn. These increased timescales both factors into the slower rate of flame propagation through the dispersed dust after ignition.

Even though the smaller particles ignite earlier and have a more rapid fireball expansion, they require more time to completely deplete. Only 60% of the Al particles for case 8 (2 μm-diameter) are consumed during the end of the rapid expansion period of the fireball. After this expansion period, the smaller particles are consumed more slowly. For example, ~15% of the 2 μm-diameter particles (case 8) remain unburned at 25 ms, while only ~5% of the 10 μm-diameter particles (case 10) remain unburned at this same time even though the particles in case 10 ignited much later. Potential explanation for this result is hypothesized in Sec. IV.

IV. DISCUSSION

A. Delayed and prompt ignition

The results presented in this paper show cases of delayed and prompt ignition of the TNT-dispersed Al particles. Experimentally, delayed ignition occurs after the explosively dispersed reactive powder is dispersed into a dust cloud.[8] The reactive particles ignite somewhere inside of the particle cloud after a time delay. The resulting flame spreads throughout the dispersed particle cloud as a turbulent dust flame. These experimental observations of the dispersal, ignition, and turbulent flame propagation for delayed ignition qualitatively match the computed results from cases 1 and 3–11. In all of these cases, they ignite and rapidly burn on delayed timescales ranging from $\mathcal{O}(5)$ to $\mathcal{O}(30\,\text{ms})$ depending on particle diameter, initial particle packing, and layer thickness. Thus, the ignition and combustion mode of the Al particles for case 1 and 3–11 represent delayed ignition.[8]

The combustion dynamics for case 2 are qualitatively different than the cases. Figure 11 shows that the heat release rate for case 2 peaks very promptly and almost immediately. The position–time diagram and time sequence shows that there are intensely burning particles at timescales far less than 1 ms. Examination of the flame structure for case 2 in Fig. 10 shows that the flame ignites at very early times. The results also show that intense burning occurs on the outside of the Al particles cloud while it radially expands. Thus, the timing and structure of the Al flame are both completely different than the other cases. If observed in an experiment, the burning Al particle cloud for case 2 would look like an expanding, white-hot fireball, which is similar to experimental images of prompt ignition of nitromethane-dispersed magnesium particles.[8] Thus, results from case 2 represent an example of prompt ignition.

B. Ignition and combustion mechanism for prompt ignition

Figure 16 shows a zoomed-in region of the Al particle volume fraction and gas-phase temperature for case 2. The Al particles are propagating into the compressed air behind the air shock. The thermodynamic conditions behind the expanding shock wave in the air are ~2500 K and 4.8 MPa with a density of 6.65 kg/m^3 at 106 μs. The Al particle volume fraction is around 1.5% near local perturbations and over 5% in cusps where particles collect. At these thermodynamic conditions, the overall equivalence ratio, ϕ, of the Al particles expanding into the shock-heated air is between 20 and 80 depending on if the location is near a cusp or near a local perturbation, as shown in Fig. 16. An ignition kernel of Al forms at 112.9 μs at the tip of a localized perturbation where the mixture is closer to the stoichiometric condition and there is less particle mass to convectively heat by the shock-heated air. (These perturbations eventually grow into the particle fingers shown in Fig. 10 by a multiphase drag-related analog

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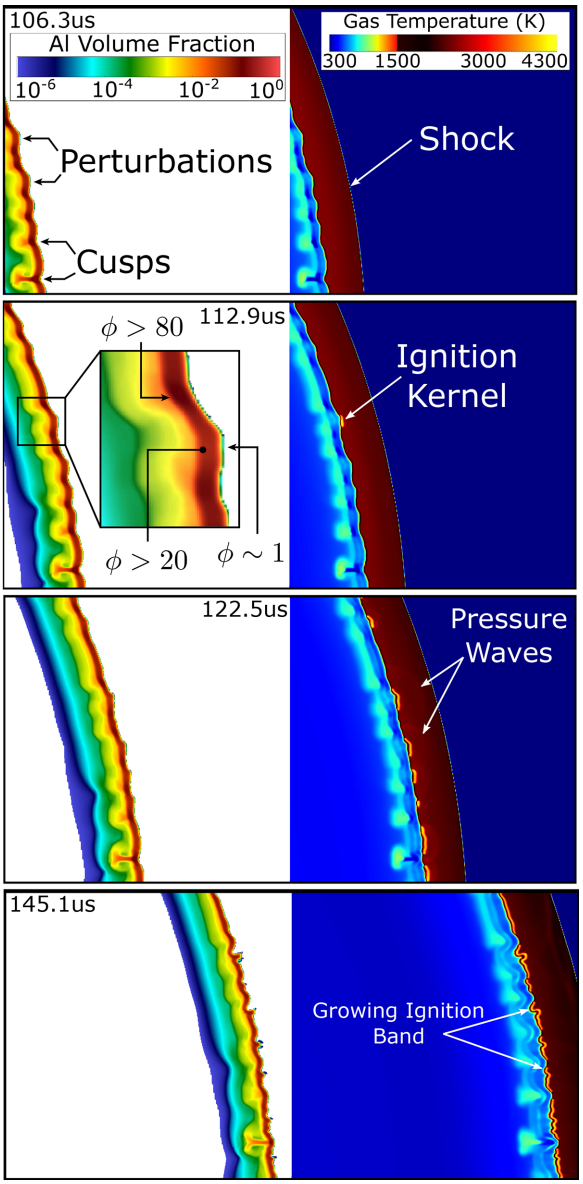


FIG. 16. Zoomed in (0.6 × 0.6 m^2) time sequence of (left) Al particle volume fraction and (right) gas-phase temperature for case 2. The time in ms is shown in the upper left-corner of each volume fraction field. Multimedia view: https://doi.org/10.1063/5.0065312.3

to baroclinic vorticity generation.[15,66]) Ignition kernels, as a result, form in the perturbation regions first. Additional flame kernels form after regions with higher volume fraction have had enough time to convectively heat energy from the shock-compressed air and ignite. After 145 μs, the entire outer edge of Al particles has ignited. The initiation of the ignition kernels produces acoustic waves that propagate to and reinforce the primary shock.

The overall flame structure shown in Figs. 10 and 16 is a non-premixed flame, where there are separate fuel-rich and oxidizer-rich regions. This is opposite to the premixed Al dust flames observed in the delayed ignition cases where the dispersed particles and air are mixed prior to ignition. Burning of the dispersing Al particles on the outermost layer does not occur with delayed ignition scenarios. The delayed ignition cases have much more particle mass that removes significantly more mechanical energy from the gas-phase shock. This lowers the post-shock-temperature to be roughly 1000 to 1500 K, which are too low to promptly ignite the particles at outer edge of the particle cloud.

Molecular diffusion is not taken into account in this model, and the flow at very early times is essentially purely gasdynamic without substantial hydrodynamic instabilities to mix the fuel and oxidizer. Thus, the traditional fuel and oxidizer mixing mechanisms for gas-phase flames based on turbulent and molecular diffusion are not active at early times in prompt ignition. However, unlike traditional gas-phase non-premixed flames, the fuel and oxidizer have different velocities in multiphase flames. Velocity slip (the relative velocity between the fuel particles and gaseous oxidizer) can introduce additional mixing.[66,67] If air is moving into the dust cloud from the reference frame of the particles, the particles at the outer edge will be exposed to fresh oxidizer and continue to burn after ignition. High-temperature products from particles burning on the outer edge will be transported further inside of the dust cloud by the gas flow. The resulting convective heat transfer from the high-temperature products increases the temperature of the particles and allows them to ignite once the they are exposed to fresh air. Similar convective enhancement effects of oxidizer and products flowing over particles have been shown to induce oscillatory combustion in expanding 1D laminar flames propagating through carbon dust clouds.[68]

Figure 17 shows the slip velocity between the gas and particles. The blue regions indicate locations where the particles are expanding radially faster than the gas, which is the majority of the expanding

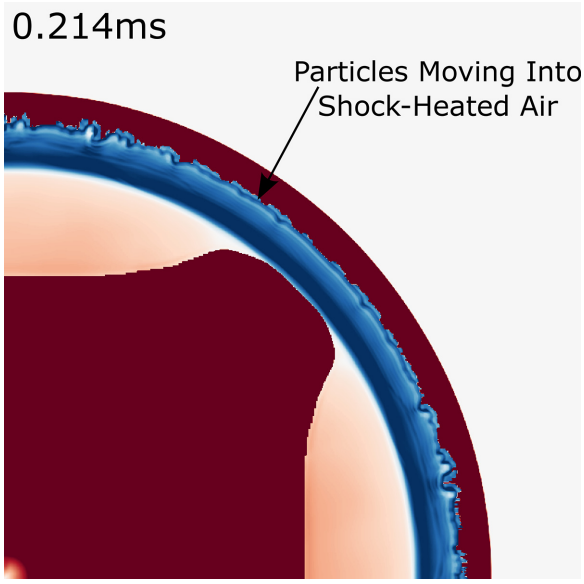


FIG. 17. Plot showing the slip velocity in case 2 (zoomed in to 0.6 × 0.6 m^2). Blue regions indicate particles expanding radially faster than gas. Red indicates gas moving faster than particles.

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Al dust cloud. In these regions, the air flow is flowing radially inward from the reference frame of the particles. Thus, mixing of the Al particles and air is by convective transport of air through the particle cloud for the initial phases of prompt ignition.

C. Potential for slip effects to influence combustion in delayed ignition

The velocity slip effects that mix fuel and oxidizer for prompt ignition scenarios are also active in cases that exhibit delayed ignition. This phenomenon potentially explains why the computed results predict robust mixing without molecular transport. We also hypothesize slip effects explain how 10 μm-diameter particles are more completely consumed before 2 μm-diameter particles as shown in Fig. 15(a) even though the smaller particles have much shorter heating and burning timescales. Fully exploring the mechanisms of that result is a topic of our ongoing work.

The rapid Al consumption phase that occurs in the delayed ignition cases occurs when the turbulent flame burns through the Al particles that have been dispersed. However, during this time the Al particles have not been perfectly mixed and are not fully dispersed as shown in the volume fraction field in Fig. 7. As a result, there are localized regions where unburned particles remain after the rapid expansion of the Al fireball. The majority of these unburned particles are located in high-temperature regions of the fireball where all the oxidizer has been consumed. If the Stokes number is small (small-diameter particles), the particles will closely follow the gas. Thus, small-diameter particles will more-or-less be trapped inside oxidizer-depleted regions of the fireball due to low velocity slip. Particles with large Stokes number (large-diameter particles) will follow the gas much less closely. The resulting slip velocity will be larger, and the Al particles will be less likely to be permanently trapped in oxidizer depleted regions. Shocks, turbulence, and hydrodynamic instabilities transport oxidizer around larger particles more rapidly due to the larger velocity slip effect. This velocity slip effect could partially explain the Al mass trends in Fig. 15. Fully exploring the influence of velocity slip on the combustion of explosively dispersed dust is a topic of our ongoing work.

V. CONCLUSIONS

In this paper, a series of two-dimensional axisymmetric numerical simulations of the ignition and combustion of an annular Al particle shell dispersed by a TNT charge were presented. The physical model fully couples the compressible reacting Euler equations of gas-dynamics to a kinetic-theory-based Eulerian granular flow model for the Al particles. The granular model accounts for compaction, collisional effects, etc. The calculations show commonly observed features of explosive particle dispersal including the formation of particle fingers and jets.

Two different types of combustion modes were observed in the computed results: delayed ignition and prompt ignition. Delayed ignition occurs in scenarios with higher Al particle mass or large particle diameter. In delayed ignition, the particles are dispersed radially outward by the expanding flow from the explosive charge. Initially, the particles are impulsively pushed away from the detonation products and TNT fireball, which are the only regions with the high temperature to ignite the particles. However, some of the particles near the inner edge of the particle cloud are transported in the reverse direction toward the TNT fireball during the negative phase of the blast. These inwardly moving particles interact with the TNT fireball and ignite. The resulting flame kernel grows and produces a turbulent premixed dust flame that propagates through the dispersed dust cloud.

Prompt ignition was observed in cases with thin annular Al particle shells with respect to the HE charge diameter. The ignition of the Al particles was rapid in prompt ignition and occurs shortly after the particle layer started to expand. The particles in these scenarios ignite on the outside of the expanding dust cloud due to heating by the shock-heated air. The flame structure for prompt ignition is a non-premixed Al dust flame where mixing occurs by velocity slip transporting air over the burning particles at early times.

The computed results show that smaller-diameter Al particles ignite more quickly and produce a more rapidly expanding turbulent Al fireball in delayed ignition scenarios. This trend can be explained by the lower heating and burning timescales for small-diameter Al particles. Paradoxically, explosively dispersed clouds of small-diameter Al particles require more time to be fully consumed than clouds of larger-diameter particles. After the turbulent flame propagates through the dispersed cloud of Al dust, there are some particles left over due to imperfect mixing. We hypothesize that smaller-diameter unburned Al particles are more likely to be trapped in oxidizer-depleted regions of the fireball because they closely follow the flow. Larger-diameter particles follow the gas less closely and are less likely to be trapped oxidizer depleted regions as shocks, turbulence, and fluid instabilities move gas around the particles. Fully examining this hypothesis, exploring the effects of realistic 3D turbulence, anaerobic Al combustion in the detonation products, open-air environments, and detailed chemical reaction kinetics are topics of our ongoing work.

ACKNOWLEDGMENTS

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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